Key Features and Formats of Digital Images

SPEGI302 - GRAPHIC USER INTERFACE DESIGN

Prepared by Niyongabo Emmanuel

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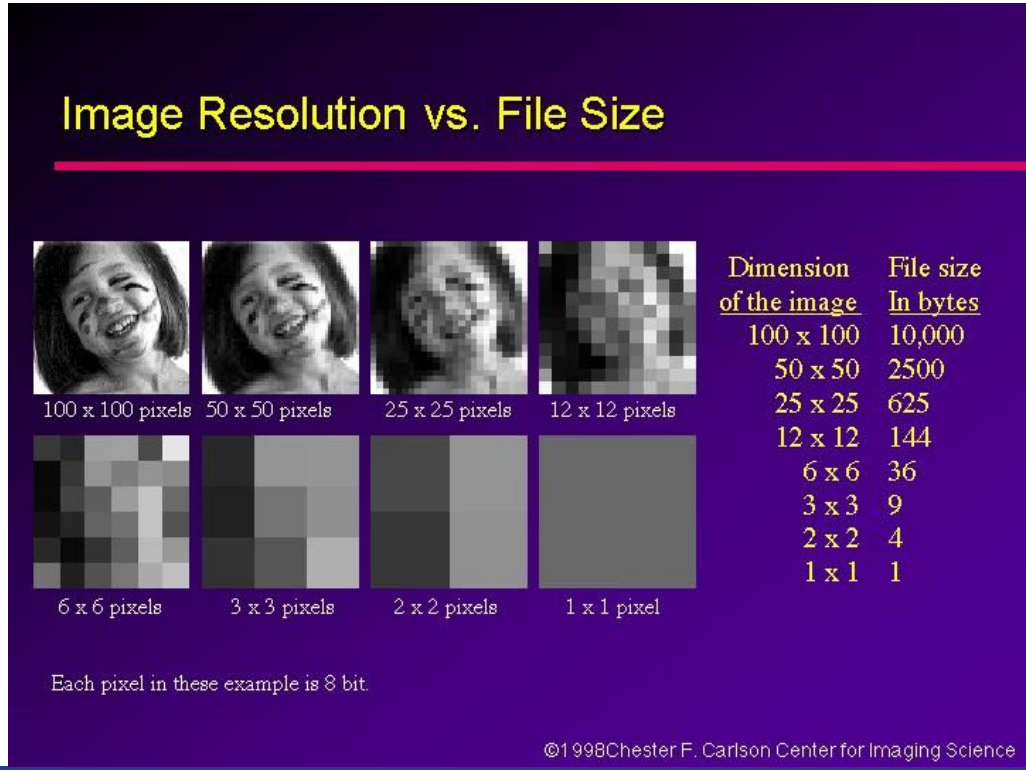
What is a Digital Image?

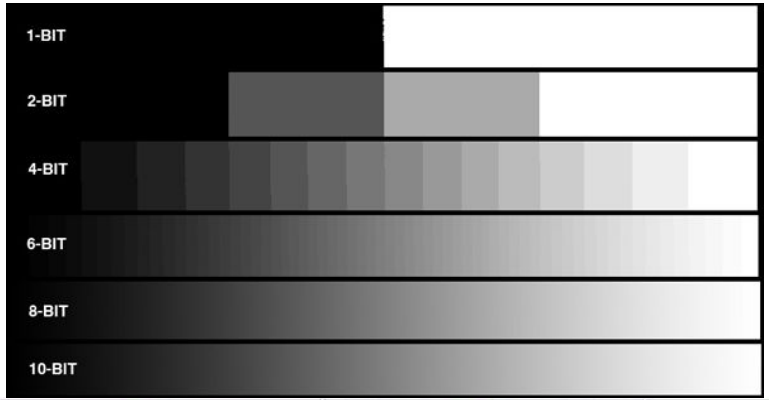
- A digital image is a computer-readable representation of a visual picture.
- Stores color, shape, and tonal information.
- Can be displayed on screens or printed

Key Features of Digital Images

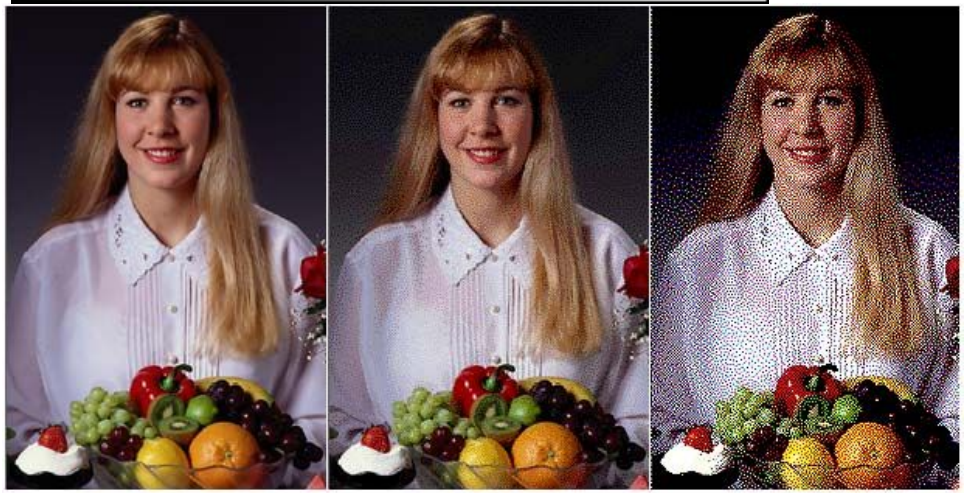
1. **Resolution (Pixels/Size)**
   - Determines detail: width × height (e.g., 1920×1080)
   - Higher resolution → more detail, larger file size
   - Enlarging raster images → pixelation



Key Features of Digital Images



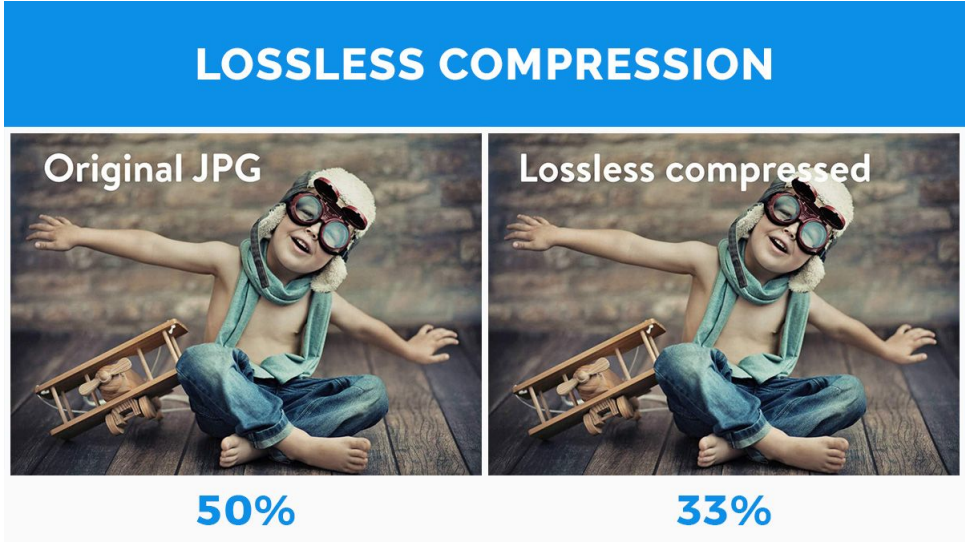
2. **Color Depth / Bit Depth**
   - Bits per pixel → number of colors
   - Examples:
     - 1-bit: black & white
     - 8-bit: 256 colors
     - 24-bit: 16.7 million colors



16.7 Million Colors | 256 Colors | 16 Colors

Compression

- **Lossless:** keeps all details (PNG) → use for logos/UI icons
- **Lossy:** removes some data to reduce size (JPEG) → use for photos/web images

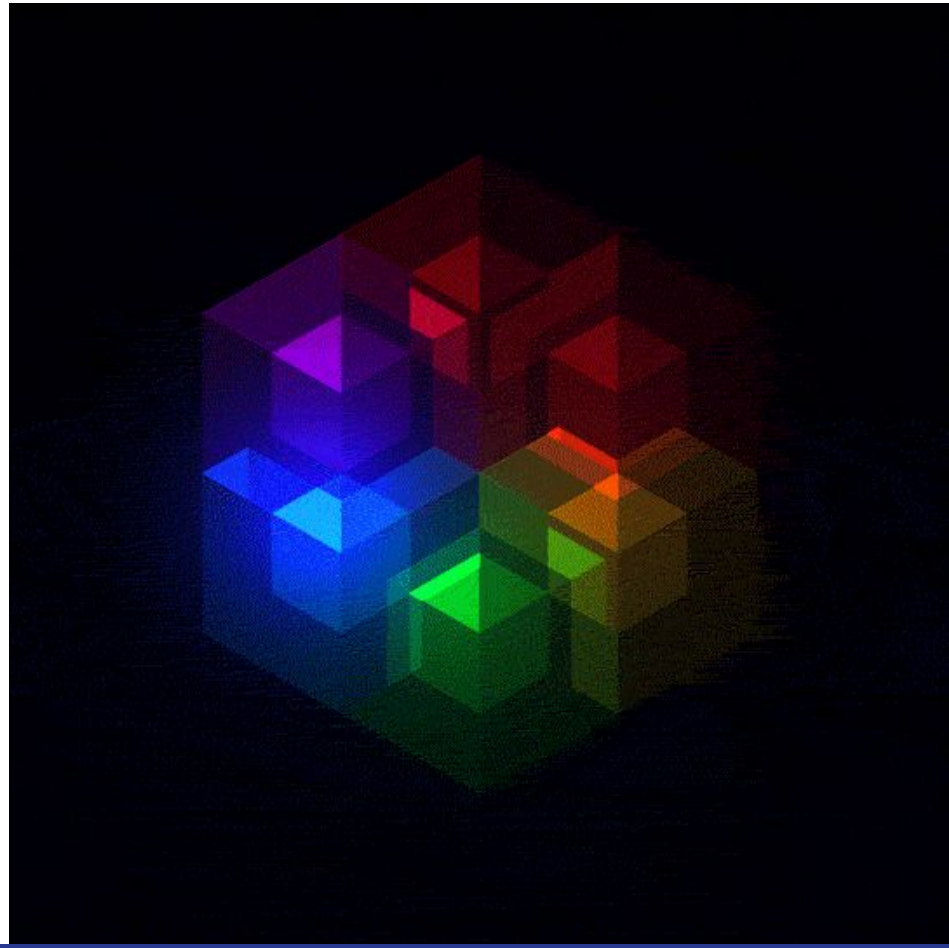


Scalability & Transparency

- **Raster:** resolution-dependent → loses quality when enlarged
- **Vector:** resolution-independent → stays sharp at any size
- **Transparency / Alpha Channel:** allows see-through areas (PNG)

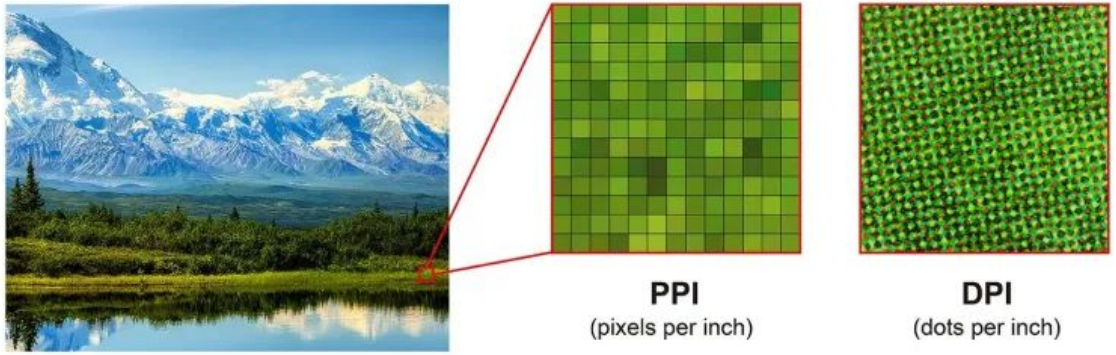
Color Space & File Size

- **RGB:** screens (digital)
- **CMYK:** printing
- File size depends on:
  - Resolution
  - Color depth
  - Compression type

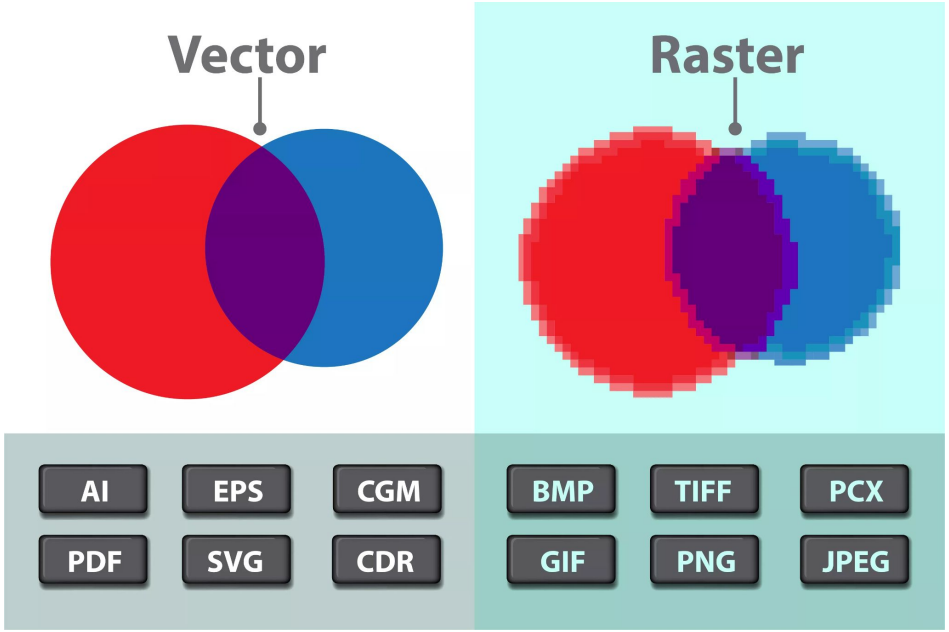


DPI (Dots Per Inch)

- Measures image sharpness / printing detail
- **Print:** 300 DPI for high quality
- **Web:** 72–150 DPI for fast loading
- Analogy: more dots → smoother, realistic image

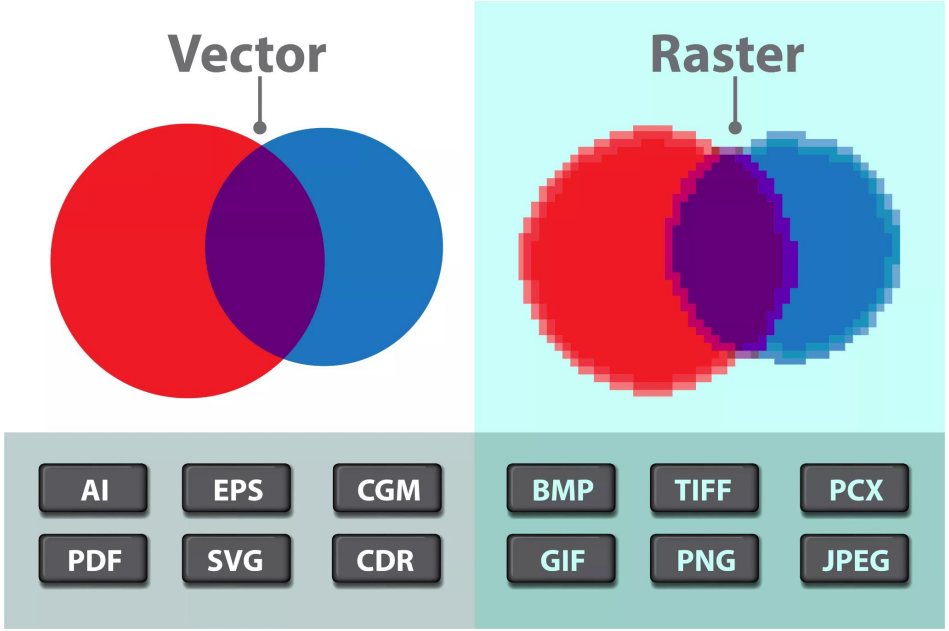


Common Raster Formats



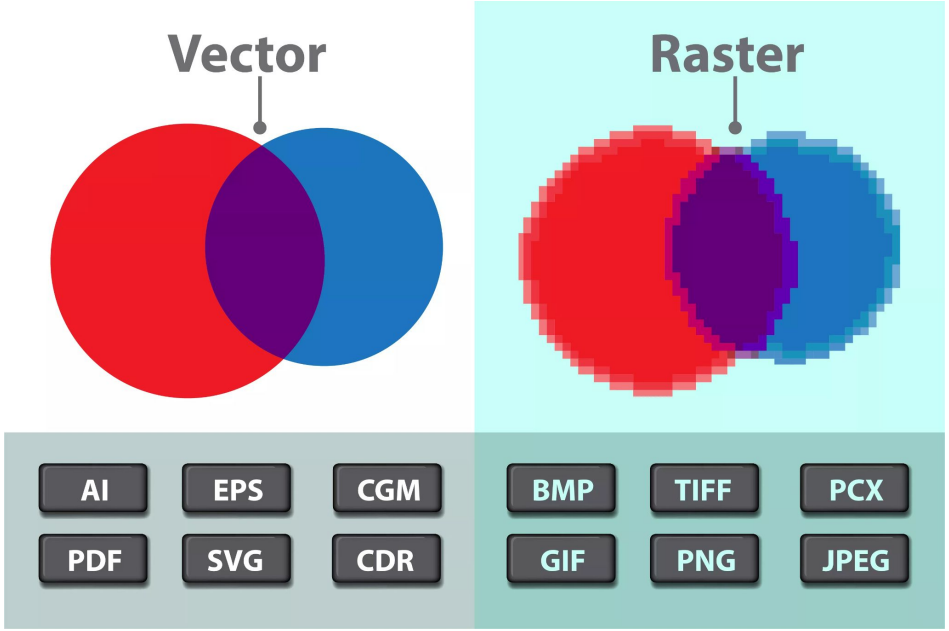
Format	Extension	Key Features	Use
JPEG	.jpg/.jpeg	Lossy, smaller	Web photos
PNG	.png	Lossless, supports transparency	Icons, UI graphics
GIF	.gif	256 colors, simple animation	Small web graphics
TIFF	.tif/.tiff	High-quality, large	Printing/archival
BMP	.bmp	Uncompressed, large	Rare/old systems
WebP	.webp	Modern, small	Web performance

Common Vector Formats



Format	Extension	Key Features	Use
SVG	.svg	XML, scalable, interactive	Web icons, responsive graphics
AI	.ai	Adobe proprietary	Professional illustration
EPS	.eps	Print-friendly	Logos, signage
PDF	.pdf	Vector + raster	Documents, print workflow

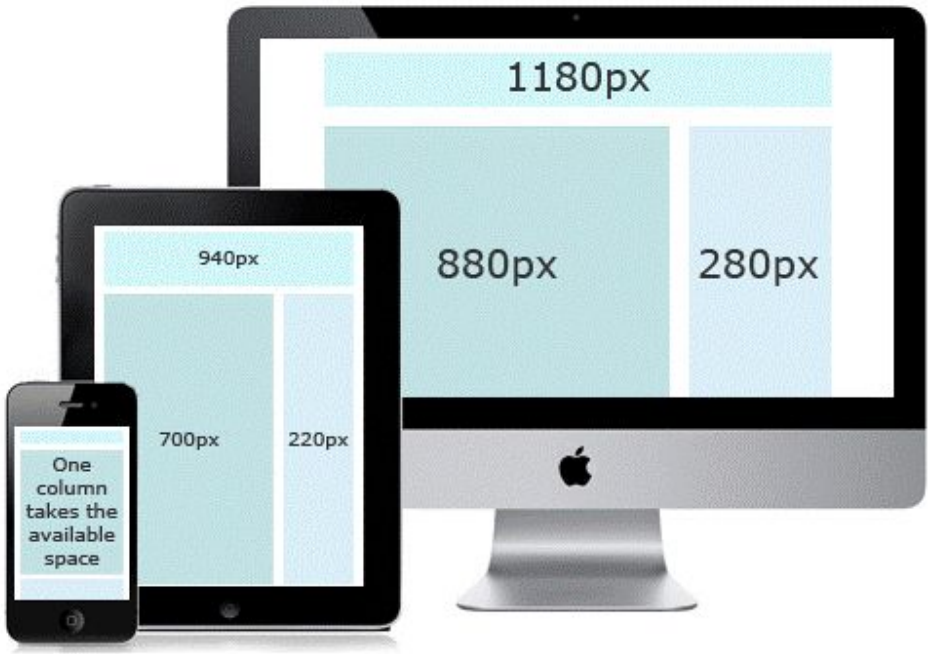
Raster vs Vector



Feature	Raster	Vector
Made of	Pixels	Paths / curves
Scaling	Loses quality	Infinitely scalable
Best for	Photos	Logos, icons, illustrations
File size	Can be large	Usually smaller
Editing	Pixel-based	Shape/line-based

Why it Matters for UI Design?

- UI must adapt to **different screen sizes** (mobile, tablet, desktop)
- Use **vector assets** for icons, logos, scalable graphics
- Use **optimized raster images** fo photos/backgrounds
- Consider **transparency and compression** for better performance



Quick Tips for Designers

- Keep **high-quality originals**; export smaller versions for devices
- Use **PNG for crisp graphics**, **JPEG for photos**
- Choose **vector for scaling**, **raster for detailed photos**
- Design for **different device resolutions** (retina, 4K, web, print)

End

Thank you!